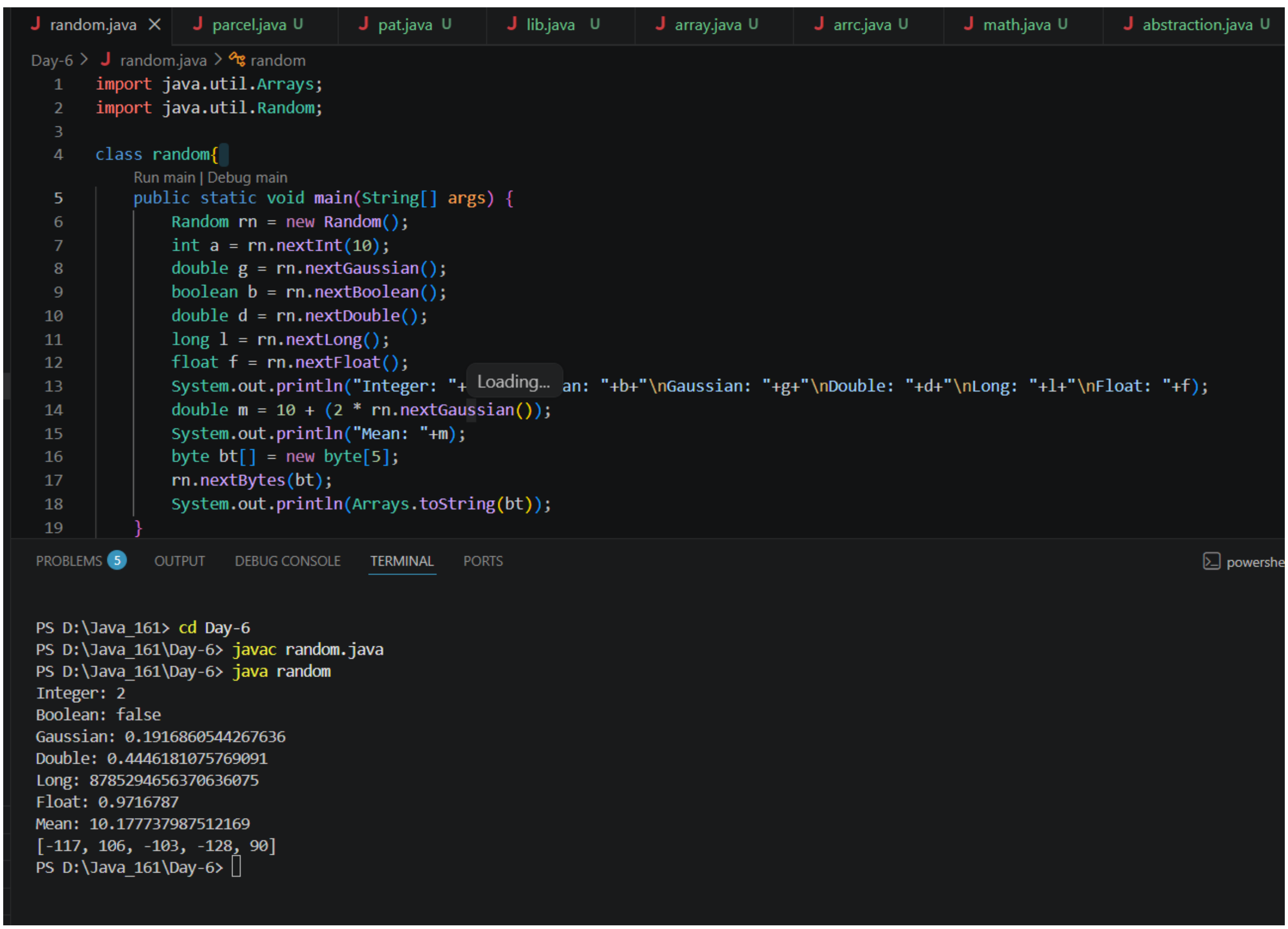
Task 1:

Write a Java program using Random to generate:

1. **Student ID** → nextInt(1000) (0–999)
2. **Marks** → nextDouble() * 100
3. **Attendance** → nextFloat() * 100
4. **Status** → nextBoolean() (Pass/Fail)
5. **Registration Number** → nextLong()
6. **Performance Score** → nextGaussian() * 10
7. **Random Bytes** → nextBytes() (array size 5)
8. **5 Random Numbers** → ints() stream



Task 2:

Write a Java program using Math class to perform the following:

1. **Find Maximum & Minimum marks** → Math.max(), Math.min()
2. **Round the marks** → Math.round()
3. **Find Square & Square Root** → Math.pow(), Math.sqrt()
4. **Find Absolute difference** → Math.abs()
5. **Find Ceiling & Floor values** → Math.ceil(), Math.floor()
6. **Find random number (0–100)** → Math.random()
7. **Find exponential value** → Math.exp()
8. **Find logarithm** → Math.log()
9. **Find sine & cosine values** → Math.sin(), Math.cos()
10. **Find cube of a number** → Math.pow(num, 3)

```java
class math{
    public static void main(String[] args) {
        //abs
        System.out.println("Absolute value: "+Math.abs(-88));
        //sqrt
        System.out.println("Square Root: "+Math.sqrt(625));
        //max
        Random rn = new Random();
        int n1 = rn.nextInt();
        int n2 = rn.nextInt();
        System.out.println("Maximum No: "+Math.max(n1,n2));
        //min
        int a1 = rn.nextInt();
        int a2 = rn.nextInt();
        System.out.println("Minimum No: "+Math.min(a1,a2));
        //round
        System.out.println("Round off: "+Math.round(8.9));
        System.out.println("Round off: "+Math.round(8.1));
        //ceil
        System.out.println("Ceil: "+Math.ceil(8.9));
        System.out.println("Ceil: "+Math.ceil(8.1));
        //floor
        System.out.println("Floor: "+Math.floor(8.9));
        System.out.println("floor: "+Math.floor(8.1));
        //cbrt
        System.out.println("Cube Root: "+Math.cbrt(1000));
        //sin,cos,tan
        Scanner s = new Scanner(System.in);
        System.out.println("Enter Angle");
        double degree = s.nextDouble();
        double rad = Math.toRadians(degree);
        System.out.println("Sin: "+Math.sin(rad)+"\nCos:"+Math.cos(rad)+"\nTan:"+Math.tan(rad));
        //pow
        int b1 = rn.nextInt();
        int b2 = rn.nextInt();
        System.out.println("Power: "+Math.pow(b1,b2));
```

```
PS D:\Java_161\Day-7> javac math.java
PS D:\Java_161\Day-7> java math
Absolute value: 88
Square Root: 25.0
Maximum No: -77251836
Minimum No: -1805449639
Round off: 9
Round off: 8
Ceil: 9.0
Ceil: 9.0
Floor: 8.0
floor: 8.0
Cube Root: 10.0
Enter Angle
1
Sin: 0.01745240643728351
Cos:0.9998476951563913
Tan:0.017455064928217585
Power: Infinity
PS D:\Java_161\Day-7>
```

Task:3

Scenario:

A bank system allows users to perform basic account operations like deposit and withdrawal using abstraction.

Task:

● Initial Balance = 0

Create an **abstract class** Account with:

- Abstract methods:
  - deposit(double amount)
  - withdraw(double amount)

Create a subclass MyAccount to:

1. Add amount to balance using deposit()

2. Subtract amount from balance using withdraw()
3. Maintain and update balance properly

Operations to Perform:

1. Deposit = 1000
2. Withdraw = 400

Expected Output:

Deposited: 1000

Withdrawn: 400

Balance: 600

```java

import java.util.Scanner;

class abstraction{
    Run main | Debug main
    public static void main(String[] args) {
        int pin_no = 1512;
        Scanner s = new Scanner(System.in);
        bank_server bs = new bank_server();
        System.out.print("Enter your pin: ");
        int pin = s.nextInt();
        int choice;
        if(pin_no != pin){
            System.out.println("Wrong Pin");
            System.exit(0);
        }
        do{
            System.out.println("Enter your choice: \n1.Check Balance \n2.Deposite \n3.withdrawl: \n4.Enter Zero to exit");
            System.out.print("Your Choice: ");
            choice = s.nextInt();
                if(choice == 1){
                    System.out.println("Balance amt: "+bs.getbalance());
                }
                else if(choice == 2){
                    System.out.print("Enter your amt: ");
                    double a = s.nextDouble();
                    bs.deposite(a);
                }
                else if(choice == 3){
                    System.out.print("Enter your amt: ");
                    double b = s.nextDouble();
                    bs.withdrawl(b);
                }

                else{
                    System.out.println("Process Completed..");
                }
```

Day-7 > abstraction.java > bank_server

```java
class abstraction{
    public static void main(String[] args) {
                else{
                    System.out.println("Process Completed..");
                }
            }while(choice!=0);

        }

    }

class bank_server{
    private double balance = 5000.7;
    double getbalance(){
        return balance;
    }
    void deposite(double amt){
        balance += amt;
    }
    void withdrawl(double amt){
        if(balance<=amt){
            System.out.println("Insufficient balance");
        }
        else
        balance -= amt;
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Power: Infinity
PS D:\Java_161\Day-7> javac abstraction.java
PS D:\Java_161\Day-7> java abstraction
Enter your pin: 1512
Enter your choice:
1.Check Balance
2.Deposite
3.withdrawl:
4.Enter Zero to exit
Your Choice: 1
Balance amt: 5000.7
```